

HydraSense Smart Unit — Product & Functional Requirements

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What the Product Is

The Smart Unit is a small electronic module that sits at the base of a water bottle. It measures how much water a person drinks, how clean the water is, and how hot or cold it is. It connects to a mobile app over Bluetooth and uses LED lights and a small buzzer to communicate with the user. All settings are managed through the app. The module runs on a battery and should work for days without needing to be charged.

PART 1 — PRODUCT REQUIREMENTS (PRD)

1. Product Goals

- Help users track how much water they drink every day without any manual effort.
- Warn users when their water quality or temperature is outside their personal comfort zone.
- Remind users to drink water at times they choose.
- Provide a simple, visual experience through LED lights and audio cues.
- Allow the same physical unit to be paired with a different user when needed.

2. User Goals

Goal	What the User Expects
Track daily water intake	The bottle automatically counts every sip, no input needed
Know if water is safe to drink	The bottle warns them if water quality is poor
Know if water is too hot	The bottle warns them before they drink dangerously hot water
Get reminded to drink	Lights and sounds nudge them based on their own schedule
See their progress	LED lights show how charged the battery is and how synced the data is
Use the bottle as a lamp	When charging, the bottle can glow as a mood light

3. Product Scope

Included in This Version

- All sensor tracking (water amount, water purity, water temperature)
- LED feedback for all situations
- Buzzer sounds for all situations
- Mobile app configuration for all user preferences
- First-time setup and registration experience
- Data storage for up to 30 days on the device itself
- Bluetooth data sync with the app
- Lamp mode when plugged into charger
- Reset options (soft, factory, physical button)

Not Included in This Version

- The mobile app itself (separate project)
- Cloud storage or server backend
- Regulatory certification

PART 2 — FUNCTIONAL REQUIREMENTS (FRD)

1. User Preferences — Settings the User Controls via App

These are the variables the user sets through the mobile app. The device stores them and uses them to decide how to behave. All settings have sensible default values so the product works immediately out of the box.

1.1 Water Purity Goal (water_purity_goal)

- The user sets a purity threshold — the level of dissolved content in water that they consider acceptable.
- If the water in the bottle exceeds this threshold, the device alerts the user with lights and a sound.
- If the user does not set this, a default value is used automatically.
- The user can change this at any time through the app. The change takes effect immediately on the device.

1.2 Water Temperature Goal (water_temp_goal)

- The user sets a temperature threshold above which they consider the water too hot to drink.
- If the water temperature exceeds this threshold, the device alerts the user.
- A default value is used if the user does not specify one.
- The user can change this at any time through the app.

1.3 Daily Hydration Goal (hydration_goal)

- The user sets how much water they aim to drink in a day, in millilitres.
- This goal is used to calculate the user's hydration score throughout the day.
- A default value is used if the user does not set one.
- The user can change this at any time.

1.4 Reminder Window and Frequency

The user sets three things:

- **Window start time** — the earliest time in the day when reminders should be shown (e.g., 8:00 AM).
- **Window end time** — the latest time when reminders should stop (e.g., 8:00 PM).
- **Reminder frequency** — how often to remind within that window (e.g., every 1 hour).

The device will only show hydration reminders within this window. Outside this window, no reminders are shown.

1.5 Hydration Reminder Color (hydration_reminder_color)

- The user can choose a custom LED color for hydration reminders.
- If a custom color is chosen, it is used **only** when the user's hydration score is above 80 (doing well).
- When the score is below 80, the standard color system is used regardless of this setting (see LED Logic section).

1.6 Hydration Reminder Sound (hydration_reminder_sound)

- The user can choose a sound from a list of available options to play during hydration reminders.
- Sounds are selected via the app.
- See Section 5 for the full list of suggested sounds.

2. Device Identity

2.1 Unique Device ID

- Every physical unit has a unique identifier (MAC address) built in at the time of manufacturing.
- This ID is included in every piece of data and every command sent over Bluetooth.
- This ensures that when two bottles are in the same room and both connected to the same phone, the app can tell them apart and send commands to the right one.

2.2 Command Metadata

Every command sent to or from the device must include:

- The device's unique ID
- The time the command was sent
- The bottle model name/type
- The current firmware version on the device

This applies to all commands: settings changes, syncs, resets, mode changes, and alerts.

3. Bluetooth Behaviour

- The device communicates with the mobile app over Bluetooth.
- It is always discoverable and ready to connect when powered on.
- When the app connects, any data that hasn't been sent yet is automatically transferred.
- If the app disconnects while a reading is being taken or an alert is active, the device continues normally and queues data for the next connection.
- The device does not need a constant Bluetooth connection to function — it works fully offline and syncs whenever the app is nearby.

4. First Time Setup & Registration Journey

This is the experience when a user takes the product out of the box for the first time.

Step 1 — Power On

- The user charges the device if the battery is empty, then powers it on.
- On first power on, the device knows it has never been set up before.
- It communicates this by playing a specific startup sound and displaying a specific LED pattern to indicate it is waiting to be registered.

Step 2 — Pairing with the App

- The user opens the app on their phone.
- The app discovers the device using Bluetooth and guides the user through registration.
- During this process, the device gives real-time LED and sound feedback so the user knows what is happening at each step (searching, connecting, registering, calibrating, done).

Step 3 — Calibration

- Once paired, the device calibrates itself.
- Calibration means the device learns the weight of the empty bottle so it knows what "zero water" looks like.
- The user is guided through this by the app, with corresponding lights and sounds on the device.
- The device stores the bottle's empty weight so it can accurately calculate how much water is inside.

Step 4 — Time Sync

- When the app first connects, it tells the device what the current time and date is.
- From that point on, the device keeps track of time on its own — it does not need to be connected to the app to know what time it is.
- This ensures all readings are timestamped correctly even when Bluetooth is not connected.

Step 5 — Ready to Use

- Once registration and calibration are complete, the device is ready.
- The user's preferences are set to defaults and can be customised through the app.
- All LED and sound cues return to normal operating mode.

Re-registration (Pairing to a New User)

- If a user wants to give the bottle to someone else or remove their account from the device, they can do a factory reset (via app or physical button).
- After factory reset, the device goes back to "first time use" mode and the new user goes through the setup flow from the beginning.
- All previously stored data is wiped from the device on factory reset.

Already Registered (Returning User)

- If the device has already been registered and is powered on, it simply continues where it left off.
- All saved preferences, stored data, and configurations are still there.
- The device does not need to pair again unless the user explicitly resets it.

5. LED Behaviour

The device uses a ring of LED lights around the bottle to communicate with the user. Every situation has a defined colour and pattern.

5.1 Hydration Reminders (During Reminder Window)

Triggered at the frequency set by the user, within their reminder window.

Hydration Score	LED Colour	Pattern
Score > 80 (On track)	Green — or user's custom colour if set	Moderate blink for 3 seconds
Score between 50–80 (Falling behind)	Amber	Moderate blink for 3 seconds
Score < 50 (Low)	Red	Moderate blink for 3 seconds

5.2 Water Purity Alert

- Triggered when water purity exceeds the user's `water_purity_goal`.
- Colour: Purple
- Pattern: Two sharp quick flashes

5.3 Temperature Alert

- Triggered when water temperature exceeds the user's `water_temp_goal`.
- Colour: Orange
- Pattern: Two sharp quick flashes

5.4 Calibration in Progress

- Shown during initial setup and recalibration.
- Colour: Amber
- Pattern: Slow flowing/wave pattern — keeps going until calibration is complete

5.5 Successful Reading Confirmation

- Shown after the device successfully records a drinking event.
- Colour: Green
- Pattern: Soft gentle pulse, 2 times

5.6 Bluetooth Sync Success

- Shown when data successfully syncs to the app.
- Colour: Cyan (light blue)
- Pattern: Quick sweep across all LEDs once

5.7 Charging — Battery Progress Bar

- Shown whenever the device is plugged into a charger.
- The LEDs act like a progress bar: the number of lit LEDs represents the current charge percentage.
- Example with 10 LEDs: if the battery is at 50%, 5 LEDs glow solid green, the 6th LED blinks slowly, and the remaining 4 are off.
- When fully charged, all LEDs glow solid green steadily.

5.8 Low Battery Warning

- Triggered when battery drops below 10%.
- Colour: Red
- Pattern: 3 slow pulses — at most once per hour so it isn't annoying

5.9 Lamp Mode

- Only available when the device is plugged into a charger.
- The user picks any colour from the app.
- Pattern: Smooth, slow flowing/river-like animation in the chosen colour.
- Lamp mode turns off automatically if the charger is unplugged.

5.10 Registration / First Time Setup

- Colour: Blue
- Pattern: Slow breathing pulse while waiting to be paired

5.11 Factory Reset Warning

- Colour: Red
- Pattern: 5 rapid flashes before reset executes (gives the user a moment to cancel)

5.12 Error / Sensor Problem

- Colour: White
- Pattern: 5 rapid flashes — shown when something goes wrong internally

6. Buzzer / Sound Behaviour

Sound behaviour, buzzer patterns, and audio cues for each device state will be defined separately. To be discussed.

7. Data Storage & Sync

7.1 What Gets Stored

Individual Drinking Events (Detailed Log):

Every time the user drinks, the device stores:

- The time the reading was taken
- How much water was consumed in that event (in ml)
- The water purity reading at that time
- The water temperature at that time

These detailed records are what allow the app to show hourly drinking patterns. For example: 200ml at 7am, 400ml at 8am, 150ml at 9am, 800ml at 11am are all stored as four separate entries.

These detailed records are kept on the device only until they are transferred to the app via Bluetooth. Once transferred, they can be cleared from the device to free up space.

Daily Summary (Kept for 30 Days):

Even after the detailed records are synced or cleared, the device keeps a daily summary for the last 30 days. Each daily summary contains:

- The date
- Total water consumed that day
- Average water purity for that day
- Average water temperature for that day

After 30 days, the oldest daily summary is removed to make space for the new day.

7.2 If the User Doesn't Connect to the App

- If a user goes a full day without connecting to the app, all individual readings from that day are safely stored on the device.
- The device keeps storing day after day until 30 days of daily summaries are full.
- The next time the app connects, all unsynced data is automatically transferred.
- No data is ever lost just because the app wasn't nearby.

7.3 Partial Fill and Refill Handling

- If the user fills the bottle when it already has water in it, the device detects the additional water poured in and counts that separately toward the day's total.
- **Example:** if the bottle has 200ml and the user adds 600ml, the device records that 600ml was added, not that 800ml appeared.
- This ensures the daily total is accurate regardless of how the user fills their bottle.

8. Reset Options

8.1 Soft Reset (Software Restart)

- Restarts the device without deleting any data or settings.
- Everything continues exactly as before — preferences, stored data, calibration, user pairing all remain intact.
- Can be triggered via the app.
- **Use case:** if the device is behaving unexpectedly and needs a quick restart.

8.2 Factory Reset (Full Wipe)

- Wipes all data, preferences, calibration, and user pairing from the device.
- The device goes back to "fresh out of the box" state.
- Requires a fresh registration and setup flow.

Can be triggered two ways:

- Through the app (with confirmation step to prevent accidents).
- Through a physical button on the device — hold for a long press (recommended: 10 seconds) to trigger. The device plays 5 warning beeps and shows red flashing lights before wiping, giving the user a chance to stop.

8.3 Best Practice Note for the Team

For the physical button, it is strongly recommended to have two distinct actions:

- **Short press** — wakes the device or confirms an action.
- **Long press (3 seconds)** — powers on/off.
- **Very long press (10 seconds)** — factory reset.

This prevents accidental resets during normal use. The reset should not execute immediately — there should be a 5-second confirmation window (with lights and sound) before it completes, giving the user a chance to release the button and cancel.

9. Lamp Mode

- Lamp mode lets the user turn the bottle's LED ring into a mood light.
- The user picks any colour they want from the app.
- The light flows in a smooth, river-like, slow-moving pattern in that colour.
- **Lamp mode can only be turned on when the device is plugged into the charger.** This prevents it from draining the battery.
- If the charger is unplugged while lamp mode is on, lamp mode turns off automatically.
- Lamp mode is manually turned on and off by the user through the app.

10. How the Hydration Score Works

The hydration score formula, thresholds, and time-of-day weighting logic will be defined separately. To be discussed.

11. Summary of All User-Configurable Settings

Setting	What It Controls	Default If Not Set
water_purity_goal	The purity threshold above which the user gets an alert	Set to a safe standard default
water_temp_goal	The temperature above which the user gets a hot-water alert	Set to a sensible safety default
hydration_goal	How much water the user wants to drink per day	Set to a standard health recommendation
reminder_window_start	When daily reminders begin	8:00 AM
reminder_window_end	When daily reminders stop	8:00 PM
reminder_frequency	How often reminders occur in the window	Every 1 hour
hydration_reminder_color	Custom LED colour for reminders when score > 80	Green (default)

Setting	What It Controls	Default If Not Set
hydration_reminder_sound	Sound played during hydration reminders	2 gentle beeps (default)
lamp_mode_color	The colour the LEDs glow in lamp mode	White

13. Bluetooth Command Reference

Every message sent between the mobile app and the device follows a standard JSON structure. This section defines all commands — including the ones already in use — plus additional commands required for a production-grade product.

How Commands Work

Every command sent from the app to the device (or from the device back to the app) must always include these base fields:

```
{
  "tx_id": "COMMAND_NAME",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:00Z",
  "firmware_version": "1.0.0",
  "bottle_model": "HydraSense-Pro-v1"
}
```

Field	What it is
tx_id	The command name — identifies what this message is asking the device to do
mac_address	The unique ID of the device — used to distinguish two bottles in the same room
timestamp	When the command was sent (UTC format)
firmware_version	The firmware version currently running on the device
bottle_model	The product model — useful for compatibility checks in future versions

Existing Commands

Time Sync — Tell the device what time it is (used on first connect and periodically)

```
{
  "tx_id": "TIMESTAMP",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:00Z",
  "firmware_version": "1.0.0",
  "bottle_model": "HydraSense-Pro-v1",
  "current_time": "2025-02-21T14:30:00Z",
}
```

```
"timezone": "Asia/Kolkata"
}
```

Set User Preferences — Send all configurable settings to the device at once

```
{
  "tx_id": "INPUT_DATA",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:00Z",
  "firmware_version": "1.0.0",
  "bottle_model": "HydraSense-Pro-v1",
  "water_purity_goal": 500,
  "water_temperature_goal": 40,
  "hydration_goal": 2000,
  "reminder_window": {
    "start_time": "08:00",
    "end_time": "22:00",
    "frequency_minutes": 60
  },
  "hydration_reminder_color": "#0000FF",
  "hydration_reminder_sound": "double_beep"
}
```

Note: hydration_reminder_sound added — was missing from the original command.
frequency_minutes replaces drinking_remainder for clarity (value is now in minutes as a number, not a string).

Calibration — Empty Bottle

```
{
  "tx_id": "CALIBRATION",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:00Z",
  "firmware_version": "1.0.0",
  "bottle_model": "HydraSense-Pro-v1",
  "stage": "empty_bottle"
}
```

Lamp Mode — Enable

```
{
  "tx_id": "LAMP_MODE",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:00Z",
  "firmware_version": "1.0.0",
  "bottle_model": "HydraSense-Pro-v1",
  "status": "enable",
  "color": "#FFFFFF"
}
```

Lamp Mode — Disable

```
{
  "tx_id": "LAMP_MODE",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:00Z",
  "firmware_version": "1.0.0",
  "bottle_model": "HydraSense-Pro-v1",
  "status": "disable"
}
```

Software Reset

```
{
  "tx_id": "SOFTWARE_RESET",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:00Z",
  "firmware_version": "1.0.0",
  "bottle_model": "HydraSense-Pro-v1",
  "status": "enable"
}
```

Read Historical Aggregates — App requests the last 30 days of daily summaries

```
{
  "tx_id": "HISTORICAL_AGGREGATES",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:00Z",
  "firmware_version": "1.0.0",
  "bottle_model": "HydraSense-Pro-v1",
  "status": "read"
}
```

Factory Reset

```
{
  "tx_id": "FACTORY_RESET",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:00Z",
  "firmware_version": "1.0.0",
  "bottle_model": "HydraSense-Pro-v1",
  "status": "enable"
}
```

New Commands — Production Requirements

Device Registration — Sent by app on first-time pairing to register this unit to a user

```
{
  "tx_id": "REGISTER_DEVICE",
  "mac_address": "D4:36:39:B6:1B:70",

```

```
"timestamp": "2025-02-21T14:30:00Z",
"firmware_version": "1.0.0",
"bottle_model": "HydraSense-Pro-v1",
"user_id": "usr_abc123",
"device_nickname": "My HydraSense"
}
```

Unpair Device — Remove this device from the current user account, prepare for fresh setup

```
{
  "tx_id": "UNPAIR_DEVICE",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:00Z",
  "firmware_version": "1.0.0",
  "bottle_model": "HydraSense-Pro-v1",
  "status": "enable"
}
```

Read Detailed Sensor Logs — App requests individual drinking event records (not just daily totals)

```
{
  "tx_id": "SENSOR_LOGS",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:00Z",
  "firmware_version": "1.0.0",
  "bottle_model": "HydraSense-Pro-v1",
  "status": "read",
  "from_date": "2025-02-20",
  "to_date": "2025-02-21"
}
```

Acknowledge Data Sync — App tells device that data was received successfully so the device can mark it as synced

```
{
  "tx_id": "SYNC_ACK",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:00Z",
  "firmware_version": "1.0.0",
  "bottle_model": "HydraSense-Pro-v1",
  "synced_up_to": "2025-02-21T14:29:00Z"
}
```

Device Status Check — App requests current health snapshot of the device

```
{
  "tx_id": "DEVICE_STATUS",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:00Z",
  "firmware_version": "1.0.0",

```

```
"bottle_model": "HydraSense-Pro-v1",
"status": "read"
}
```

Device responds with:

```
{
  "tx_id": "DEVICE_STATUS",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:05Z",
  "firmware_version": "1.0.0",
  "bottle_model": "HydraSense-Pro-v1",
  "battery_percent": 72,
  "is_charging": false,
  "sensor_temp_ok": true,
  "sensor_tds_ok": true,
  "sensor_weight_ok": true,
  "storage_used_pct": 14,
  "last_sync": "2025-02-20T22:00:00Z",
  "is_calibrated": true,
  "is_registered": true
}
```

Ping / Heartbeat— Simple check to confirm the device is alive and connected

```
{
  "tx_id": "PING",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:00Z",
  "firmware_version": "1.0.0",
  "bottle_model": "HydraSense-Pro-v1"
}
```

Device responds with:

```
{
  "tx_id": "PONG",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:00Z",
  "firmware_version": "1.0.0",
  "bottle_model": "HydraSense-Pro-v1"
}
```

Read Current Preferences — App asks device to return what settings are currently stored on it

```
{
  "tx_id": "GET_CONFIG",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:00Z",
  "firmware_version": "1.0.0",

```

```
{
  "bottle_model": "HydraSense-Pro-v1",
  "status": "read"
}
```

Error Log — App requests the device's internal error history (sensor failures, crashes, etc.)

```
{
  "tx_id": "ERROR_LOG",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:00Z",
  "firmware_version": "1.0.0",
  "bottle_model": "HydraSense-Pro-v1",
  "status": "read"
}
```

Firmware Update Notification — App tells device a new firmware version is available and initiates OTA update

```
{
  "tx_id": "FIRMWARE_UPDATE",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:00Z",
  "firmware_version": "1.0.0",
  "bottle_model": "HydraSense-Pro-v1",
  "new_version": "1.1.0",
  "status": "initiate"
}
```

Standard Device Response (ACK)

For every command the app sends, the device should respond with a simple acknowledgment so the app knows the command was received and acted on:

```
{
  "tx_id": "ACK",
  "in_response_to": "INPUT_DATA",
  "mac_address": "D4:36:39:B6:1B:70",
  "timestamp": "2025-02-21T14:30:01Z",
  "firmware_version": "1.0.0",
  "bottle_model": "HydraSense-Pro-v1",
  "status": "success",
  "message": "Preferences updated successfully"
}
```

If a command fails, the device responds with:

```
{
  "tx_id": "ACK",
  "in_response_to": "INPUT_DATA",
```



```

"mac_address": "D4:36:39:B6:1B:70",
"timestamp": "2025-02-21T14:30:01Z",
"firmware_version": "1.0.0",
"bottle_model": "HydraSense-Pro-v1",
"status": "error",
"error_code": "INVALID_VALUE",
"message": "water_purity_goal must be between 0 and 1500"
}

```

Full Command List

Command	Direction	Purpose
TIMESTAMP	App → Device	Sync current time and timezone
INPUT_DATA	App → Device	Set all user preferences at once
CALIBRATION	App → Device	Run calibration (empty or filled stage)
LAMP_MODE	App → Device	Enable or disable lamp mode with color
SOFTWARE_RESET	App → Device	Restart device without wiping data
FACTORY_RESET	App → Device	Wipe all data and return to first-use state
HISTORICAL_AGGREGATES	App → Device	Read 30-day daily summary data
REGISTER_DEVICE	App → Device	Pair device to a user account on first use
UNPAIR_DEVICE	App → Device	Remove user from device, prepare for new user
SENSOR_LOGS	App → Device	Read individual drinking event records
SYNC_ACK	App → Device	Confirm data was received; device marks it synced
DEVICE_STATUS	App → Device	Request current health snapshot of device
GET_CONFIG	App → Device	Read currently stored preferences from device
ERROR_LOG	App → Device	Read device's internal error history
FIRMWARE_UPDATE	App → Device	Notify device of available firmware update
PING	App → Device	Check if device is alive
PONG	Device → App	Response to PING
ACK	Device → App	Acknowledgment for every command received

End of Document — HydraSense Smart Unit PRD & FRD v2.0 (Plain English)